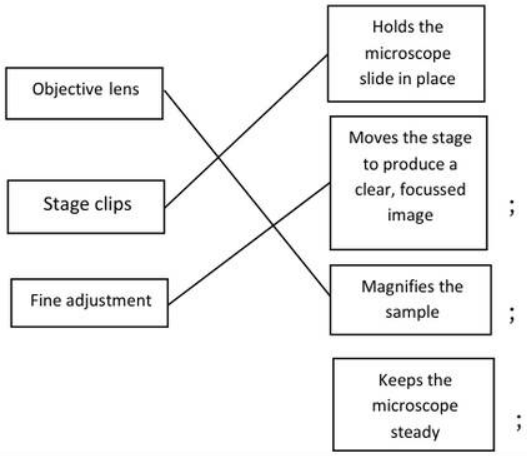
Type MS · Identify bacterial cell structures and compare with human cells

Answer 1 of 9 · Question 1 part a.ii · 2 marks

2024 JUN · Paper 01 (regional R) · find part a.ii in this mark scheme extract

Question number				
1	(a) (i)	**Any two from:** • (both) have a cell membrane; • (both) have cytoplasm; • (both) have genetic material/DNA; • (both) contain ribosomes;		(2)
	(a) (ii)	**Any two from:** • bacterial cell has a cell wall / an animal doesn't; • bacterial cell has no nucleus /an animal cell does; • bacterial cell has a plasmid /the animal cell doesn't; • bacterial cell has a flagellum/ the animal cell doesn't; • bacterial cell has pili /the animal cell doesn't; • bacterial cell has no mitochondria /the animal cell does; • bacterial cell has slime capsule / the animal cell doesn't;		(2)
	(b) (i)	 ; ; ;	Reject more than one line from each part	(3)
	(b) (ii)	• substitution, 5 ÷ 50 000 = 0.0001(cm); • conversion to mm - 0.0001 x 10; • 0.001 mm/1×10^{-3}; or • 5×10=50mm; • 50÷50000; • 0.001mm/1×10^{-3}	ecf for mp 2 and 3 Full marks for correct final answer	(3)

Total for Question 1 = 10 marks

Type MS · Identify bacterial cell structures and compare with human cells

Answer 2 of 9 · Question 2 part b · 3 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

2	(a) (i)			1
	(ii)	In the following order only: phagocytes; enzymes;		 1 1
	(iii)	produce antibodies/release antitoxins;		1
	(b)	<table><tr><th>Bacterial cell</th><th>Human skin cell</th></tr><tr><td></td><td>✓</td></tr><tr><td>✓</td><td>✓</td></tr><tr><td></td><td></td></tr><tr><td>✓</td><td></td></tr></table>	One mark for each correct row	3
	(c)	**D** (a single-stranded helix containing the bases AUGC); A is incorrect as RNA is not a double-stranded helix B is incorrect as RNA is not a double-stranded helix C is incorrect as RNA does not contain the base T		1

Total question 2 = 8 marks

Type MS · Identify bacterial cell structures and compare with human cells

Answer 2 of 9 · Question 2 part b · 3 marks

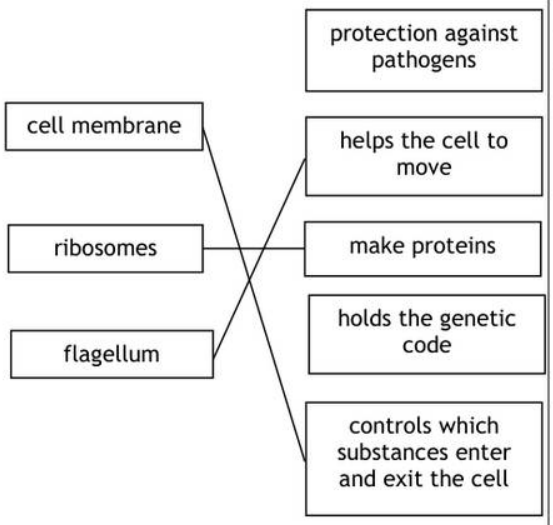
2019 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify bacterial cell structures and compare with human cells

Answer 3 of 9 · Question 3 part a.i · 3 marks

2020 JAN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

3 (a) (i)	**Structure** / **Function**  	Reject more than one line from each structure One mark for each correct line	3
(ii)	Any two from: • (bacterial cell) has a cell wall (1) • (bacterial cell) does not have a nucleus (1) • (bacterial cell) has a plasmid (1)		2
(b) (i)	**The only correct answer is B diffusion** *A is incorrect because it is down a diffusion gradient* *C is incorrect because it is not removal from the bacterium* *D is incorrect because osmosis only involves movement of water*		1
(ii)	• glucose (1) • carbon dioxide (1)	Answers must be in the correct order Allow correct chemical formulae	2
(iii)	Any two from: • to provide energy (1) • for movement (1) • for growth/reproduction/divide/mitosis (1) • for chemical reactions/producing organelles/named organelle (1)		2

Type MS · Identify bacterial cell structures and compare with human cells

Answer 3 of 9 · Question 3 part a.i · 3 marks

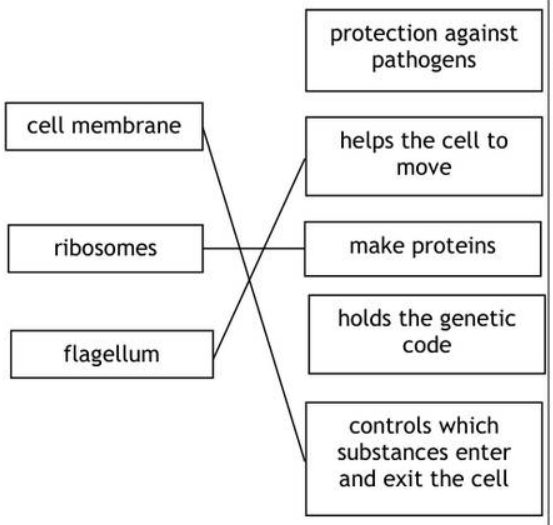
2020 JAN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Total for Question 3 = 10 marks

Type MS · Identify bacterial cell structures and compare with human cells

Answer 4 of 9 · Question 3 part a.ii · 2 marks

2020 JAN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

3 (a) (i)	**Structure** / **Function**  	Reject more than one line from each structure One mark for each correct line	3
(ii)	Any two from: • (bacterial cell) has a cell wall (1) • (bacterial cell) does not have a nucleus (1) • (bacterial cell) has a plasmid (1)		2
(b) (i)	**The only correct answer is B diffusion** *A is incorrect because it is down a diffusion gradient* *C is incorrect because it is not removal from the bacterium* *D is incorrect because osmosis only involves movement of water*		1
(ii)	• glucose (1) • carbon dioxide (1)	Answers must be in the correct order Allow correct chemical formulae	2
(iii)	Any two from: • to provide energy (1) • for movement (1) • for growth/reproduction/divide/mitosis (1) • for chemical reactions/producing organelles/named organelle (1)		2

Type MS · Identify bacterial cell structures and compare with human cells

Answer 4 of 9 · Question 3 part a.ii · 2 marks

2020 JAN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Total for Question 3 = 10 marks

Type MS · Identify bacterial cell structures and compare with human cells

Answer 5 of 9 · Question 4 part a.i · 1 mark

2023 JAN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

Number					
4	(a)	(i)	cell wall/capsule/slime coat;		1
		(ii)	55 ÷ 30 000 = 0.0018: 1.8/1.83/1.833;		2
	(b)	(i)	more densely/higher populated/more people/overcrowded/more waste/rubbish/sewage;		1
		(ii)	• fly lands on faeces (of person with a stomach ulcer); • transfers the bacteria to food; • person ingests the contaminated food;		3
	(c)	(i)	Two from: • high/medium levels in less well developed/poorer countries; • where sewage treatment is less effective/water not clean/poor sanitation; • more densely/highly populated; • healthcare/vaccines to treat infected people is less effective/available;	accept reverse arguments	2
		(ii)	Five from: • bacteria mutate/acquire resistant plasmids; • people do not finish the course/over-use of antibiotics; • resistant bacteria survive (and non-resistant bacteria die); • (resistant bacteria) reproduce/multiply; • offspring are antibiotic resistant/not affected by antibiotics; • antibiotic that will be effective is needed/used; • more serious infection occurs;		5

Total 14 marks

Type MS · Identify bacterial cell structures and compare with human cells

Answer 5 of 9 · Question 4 part a.i · 1 mark

2023 JAN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify bacterial cell structures and compare with human cells

Answer 6 of 9 · Question 8 part a.i · 3 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

Number					
8	(a)	(i)	A = flagellum; B = cell wall/capsule; C = nucleic acid/genetic material/bacterial chromosome/(loop of) DNA;		3 (grad)
		(ii)	pathogen;		1 (cler)
	(b)	(i)	• produced by microorganism/bacteria/fungi; • kills bacteria;		2 (grad)
		(ii)	A *no mark for giving X* • because larger clear zone; • antibiotic diffused from filter paper; • bacteria have been killed;		3 (exp)
		(iii)	• filter paper without antibiotic; • bacterial growth unaffected/no clear zones; • if antibiotic is causing effect;		3 (exp)

Total 12 marks

Type MS · Identify bacterial cell structures and compare with human cells

Answer 6 of 9 · Question 8 part a.i · 3 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

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Type MS · Identify bacterial cell structures and compare with human cells

Answer 7 of 9 · Question 1 part a.ii · 1 mark

2023 JUN · Paper 02 (regional R) · find part a.ii in this mark scheme extract

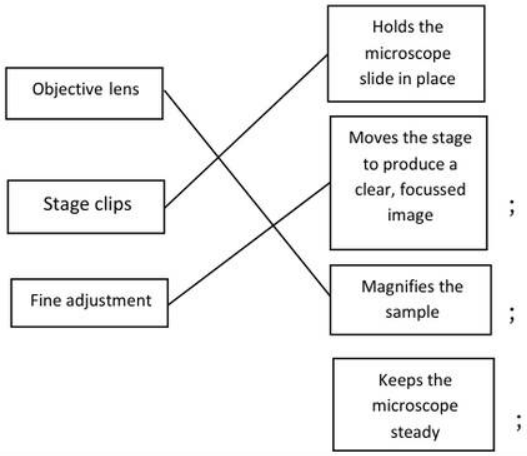
Number				
1 (a)	(i)	<table><tr><th rowspan="2">cell structure</th><th colspan="3">Function</th></tr><tr><th>makes proteins</th><th>contains DNA</th><th>allows substances in and out of cells</th></tr><tr><td>nucleus</td><td></td><td>✓</td><td></td></tr><tr><td>ribosomes</td><td>✓</td><td></td><td></td></tr><tr><td>cell membrane</td><td></td><td></td><td>✓</td></tr></table>	One mark for each correct row	3
	(ii)	cell membrane/cell wall/cytoplasm/plasmid/DNA/ribosomes/flagellum;		1
(b)	(i)	Part W phosphate (group); Part X sugar/ deoxyribose sugar;		2
	(ii)	• complementary base pairing; • A-T, G-C; • hydrogen bonds between bases;		3
(c)		153 ÷ 3; 51;	Full marks for correct final answer Ecf allow correct final answer to calculation for 1 mark	2

Total for Question 1 = 11 Marks

Type MS · Identify bacterial cell structures and compare with human cells

Answer 8 of 9 · Question 1 part a.i · 2 marks

2024 JUN · Paper 01 (regional R) · find part a.i in this mark scheme extract

1 (a) (i)	**Any two from:** • (both) have a cell membrane; • (both) have cytoplasm; • (both) have genetic material/DNA; • (both) contain ribosomes;		(2)
(a) (ii)	**Any two from:** • bacterial cell has a cell wall / an animal doesn't; • bacterial cell has no nucleus /an animal cell does; • bacterial cell has a plasmid /the animal cell doesn't; • bacterial cell has a flagellum/ the animal cell doesn't; • bacterial cell has pili /the animal cell doesn't; • bacterial cell has no mitochondria /the animal cell does; • bacterial cell has slime capsule / the animal cell doesn't;		(2)
(b) (i)		Reject more than one line from each part	(3)
(b) (ii)	• substitution, 5 ÷ 50 000 = 0.0001(cm); • conversion to mm - 0.0001 x 10; • 0.001 mm/1×10^{-3}; or • 5×10=50mm; • 50÷50000; • 0.001mm/1×10^{-3}	ecf for mp 2 and 3 Full marks for correct final answer	(3)

Total for Question 1 = 10 marks

Type MS · Identify bacterial cell structures and compare with human cells

Answer 9 of 9 · Question 4 part b.i · 3 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

4 (a)		disease causing organism/micro-organism;		1
(b)	(i)	any three from • cytoplasm; • cell/plasma membrane; • cell wall; • (loop of) nucleic acid/DNA; • flagellum;	if more than three parts labelled minus one mark for each incorrect label	3
	(ii)	• distance between A and B = 64 mm/6.4 cm; • 64 × 1000 = 64 000 µm; • magnification = $\frac{64\,000}{2}$ = 32 000;	ecf allow +/- 1 mm	3
(c)	(i)	• suitable scale, axes labelled with units; • correct way round; • points plotted correctly; • lines joining plotted points; • key to identify lines;	needs to cover half of paper award mp1,2 and 5 if bar chart plotted	5
	(ii)	any three from • both fall/vaccine becomes less effective over time; • 45-64 years more effective (over all months)/over 65 less effective (over all months); • slowly for first two months; • more/ rapid /steep decrease, between 3-4 months/after month 3 (for over 65);		3
				15 marks

Type MS · Identify bacterial cell structures and compare with human cells

Answer 9 of 9 · Question 4 part b.i · 3 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks